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Proceedings of the Combustion Institute 33 (2011) 99–123
Proceedings of the Combustion Institute
www.elsevier.com/locate/proci

# Petascale direct numerical simulation of turbulent combustion—fundamental insights towards predictive models

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**Abstract**

The advent of petascale computing applied to direct numerical simulation (DNS) of turbulent combustion has transformed our ability to interrogate fine-grained 'turbulence-chemistry' interactions in canonical and laboratory configurations. In particular, three-dimensional DNS, at moderate Reynolds numbers and with complex chemistry, is providing unprecedented levels of detail to isolate and reveal fundamental causal relationships between turbulence, mixing and reaction. This information is leading to new physical insight, providing benchmark data for assessing model assumptions, suggesting new closure hypotheses, and providing interpretation of statistics obtained from lower-dimensional measurements. In this paper the various roles of petascale DNS are illustrated through selected examples related to lifted flame stabilization, premixed and stratified flame propagation in intense turbulence, and extinction and reignition in turbulent non-premixed jet flames. Extending the DNS envelope to higher Reynolds numbers, higher pressures, and greater chemical complexity will require exascale computing in the next decade. The future outlook of DNS in terms of challenges and opportunities in this regard are addressed.
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*Keywords:* Direct Numerical Simulation (DNS); Turbulent; Combustion Models; High-Performance Computing; Complex Chemistry

## 1. Background and significance

### *1.1. Direct numerical simulation computational approach*

The rapid growth in computational capabilities in the past two decades has presented both opportunities and challenges for high-fidelity simulations of turbulent reacting flows [1–8]. The advent of petascale computing power made it possible to glean fundamental physical insight into fine-grained 'chemistry-turbulence' interactions in canonical and laboratory-scale turbulent flames with detailed chemistry using three-dimensional direct numerical simulations [9–15]. In direct numerical simulation (DNS), the instantaneous governing equations are solved without averaging or filtering; all relevant continuum scales are resolved on the grid with no closure models for turbulence or combustion using accurate numerical methods. Constitutive models representing chemical kinetics, molecular transport, and other physics are still required. Such simulations are costly, requiring tens of million of cpu-hours on a petascale computer, containing up to several

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1540-7489/$ - see front matter © 2010 Published by Elsevier Inc. on behalf of The Combustion Institute.
doi:10.1016/j.proci.2010.09.012

billion grid points, and generating 100's of terabytes of raw data. High performance computing at the petascale and beyond is enabling direct simulation of a large dynamic range of scales ($O(10^4)$ for incompressible turbulence) and the transport of dozens of species representative of hydrocarbon fuels of practical interest. The ability to combine turbulent transport with complex chemistry in DNS is providing greater realism and quantitative predictability, complementing experimental measurements in providing information critical to understanding new combustion regimes characteristic of future internal combustion engines and gas turbines for power and transport.

### *1.2. Direct numerical simulation of relevant archetypical configurations*

Next-generation alternative-fuel internal combustion engines will operate in unconventional, mixed-mode, turbulent combustion under previously unexplored 'turbulence-chemistry' regimes. Compared to current engines, combustion in next-generation engines are likely to operate at higher pressures, lower temperatures, and higher levels of dilution. Similarly, in gas turbines for power generation hydrogen-enriched fuels are being considered for pre-combustion capture of $CO_2$ to minimize greenhouse gases from fossil fuel power plants. A hydrogen-rich fuel supply poses challenges downstream of fuel delivery in terms of thermal efficiency, emissions, and flashback safety. Combustion processes in these new environments, combined with diverse physical and chemical fuel properties associated with non-petroleum based fuels, may result in complex interactions that are poorly understood even at a fundamental level.

Therefore, there is an increasing role for high-fidelity simulation approaches that capture these fundamental fine-scale turbulence-chemistry interactions, and in particular, capture and discriminate the effects of variations in fuel composition. DNS can uniquely isolate and reveal fundamental causal relationships between turbulence, mixing, and reaction that are required for physical understanding and to develop predictive models. With peta-scale computing many turbulence-chemistry interactions have recently become accessible by DNS, for example, – stabilization of autoignitive lifted turbulent jet flames, stabilization of reactive turbulent fuel jets in crossflow, premixed flame propagation, structure, and NO formation in intense turbulence, extinction and reignition in turbulent jet flames, and stratified flame propagation and structure in turbulent flames. In addition to gleaning fundamental insight into specific 'turbulence-chemistry' interactions, the DNS benchmark data are increasingly being used to validate and develop models for turbulent combustion. In recent years, the emphasis in modeling formalisms has shifted towards capturing multi-regime or mixed-mode combustion, i.e. not purely premixed or non-premixed combustion, and often propagating flames or ignition fronts through dilute, autoigniting mixtures. Because of the increased prevalence of finite-rate kinetic effects, small-scale mixing and reactive chemical scales play a more significant role in determining macroscopic combustion quantities e.g. the overall burning rate and emissions. Hence, this imposes more stringent requirements of models to capture detailed chemical effects. In this regard, DNS with complex chemistry is complementing experiments in providing essential spatially and temporally resolved three-dimensional reactive scalar and velocity field data for model evaluation. Finally, DNS data have also been used to aid in the interpretation of measurements by relating scalar statistics obtained from lower-dimensional measurements to those of the true three-dimensional quantity.

In the following sections the governing Navier–Stokes, energy and species continuity equations and constitutive models for molecular transport are presented for both compressible and low-Mach number DNS formulations together with numerical algorithms, boundary conditions, parallel domain decomposition, chemical reduction strategies, and some potentially useful tricks-of-the-DNS-trade. Next, the various roles of large-scale DNS are illustrated by selected examples, loosely organized by phenomena: stabilization of autoignitive lifted turbulent jet flames, stabilization of reactive turbulent fuel jets in crossflow, premixed flame propagation, structure, and NO formation in intense turbulence, stratified flame propagation and structure in turbulent flames, and extinction and reignition in turbulent jet flames. In the examples, implications for model formalisms that warrant further development are discussed in light of the greater significance placed on finite-rate chemical kinetics interactions with turbulence and mixed regimes of combustion. Finally, the future outlook of DNS is discussed in light of the changing landscape of high performance computing in its expected evolution from petascale to exascale supercomputers in the next decade. In particular, lessons learned from recent refactorization of a compressible Navier–Stokes solver for an existing hybrid architecture utilizing cell broadband engines are discussed, and examples of in situ feature extraction/tracking and visualization methods for addressing the data deluge accompanying large-scale simulations are presented.

There are many review articles related to advances in DNS of turbulent combustion [1,6–8,15–17]. The present intent is to illustrate the roles of large-scale 3D DNS focusing on turbulence-chemistry interactions rather than to provide a comprehensive review of the DNS literature. As such, many important multi-physics phenomena – thermal radiation, soot and multiphase flows – are beyond the present scope.

### 1.3. Compressible DNS formulation

In the compressible DNS formulation the full compressible Navier–Stokes, total energy, species and mass continuity equations coupled with detailed chemistry are typically solved with higher-order accurate, non-dissipative numerical schemes [18]. The governing equations are solved on a conventional three-dimensional structured mesh. Spatial differentiation is achieved through either high-order centered finite differences, for example, eighth-order finite differences along with tenth-order filters [19] to damp any spurious oscillations in the solution, or through compact Pade finite difference schemes [20]. Time advancement is typically achieved through explicit Runge–Kutta integration, for example, a six-stage, fourth-order explicit Runge–Kutta (R–K) method [21]. Navier–Stokes characteristic boundary condition (NSCBC) treatment [6,22–24] is typically used on the boundaries. The coupling of high-order finite difference methods with explicit R–K time integration to solve for a large system of coupled partial differential equations has traditionally been able to make very effective use of the available computing resources, obtaining spectral-like spatial resolution without excessive communication overheads and enabling scalable parallelism. The dominant target science for compressible DNS codes has been research of combustion phenomenon under intense turbulence conditions where the resolution requirements of the flame and flow are the same and uniform across a significant extent of the computational domain. In contrast, an adaptive mesh refinement (AMR) approach is more suitable for simulating flames where there is a wide disparity in scales and higher resolution is required in only a small fraction of the total domain, for example, if the flame is much thinner than the smallest turbulence scales.

Most compressible DNS codes are written entirely in fortran and are parallelized using a three-dimensional domain decomposition and MPI communication. Each MPI process is responsible for a piece of the three-dimensional domain. All MPI processes have the same number of grid points and the same computational load. Inter-processor communication is between nearest neighbors in a three-dimensional topology for centered finite difference methods. A ghost-zone is constructed at the processor boundaries by non-blocking MPI sends and receives among the nearest neighbors in the three-dimensional processor topology. The amount of data exchanged between neighbors is determined by the size of the finite difference stencil and is ~15% of the data held by each process. Global communications are not a bottleneck and are mainly required for monitoring and synchronization ahead of I/O.

The DNS codes must scale well in the weak sense to be able to run on the full extent of the largest supercomputers, typically ~50,000–224,000 processors on a Cray XT5. The most expensive part of the DNS (30–60% of the total time) is evaluation of the chemical source terms and thermochemical and transport properties. Fully coupled mass conservation equations for the different chemical species are solved as part of the simulation to obtain the chemical state of the system. Detailed and reduced chemical kinetics and mixture-averaged molecular transport models are used. Optimized routines to compute the chemical reaction and species diffusion rates based on Sandia's Chemkin package [25] are generated by parsing standard Chemkin thermodynamic and chemical mechanism input files. The autogenerated routines to evaluate thermodynamic properties (enthalpy and Gibbs free energy) or chemical reaction rates are output in a variety of languages (currently fully unrolled FORTRAN-77, FORTRAN-90 array syntax, C, and a CUDA option is under development). This automated code generation capability for the most computationally expensive kernel in the simulation provides a great deal of flexibility when moving between architectures and compilers. It also allows DNS developers to capture mechanism-specific practices that result in code optimization (for example, structuring a particular chemical mechanism routine to encourage the compiler to generate SSE vector instructions) and to be able to adapt such optimizations to other mechanisms. Furthermore, collaborations with computer scientists have demonstrated the performance advantages of graphics processors [26] for DNS with complex chemistry.

For complex chemical mechanisms ($C_2$ hydrocarbons and higher), reduced chemical and transport models that are fine-tuned to the target DNS problem are developed as a pre-processing step. During this step, the reaction state at the chosen condition is sampled from lower-dimensional representative combustion configurations such as homogeneous auto-ignition and laminar strained and unstrained flames. More recently, comparisons between unsteady laminar axisymmetric counterflow experiments and DNS have been made of transient autoignition and transient extinction/reignition dynamics in both heated and unheated oxidizer streams [27–29]. These comparisons of laminar building-block flows representative of a turbulent flame provide a more stringent validation of kinetics coupled with transport. Another interesting and potentially relevant configuration would be to measure the flame speed of a reactive flame front propagating into an autoignitive, reacting mixture. In the systematic reduction of a detailed mechanism the chemical state is first analyzed to identify the unimportant reactions and species that can be eliminated without loss of accuracy [30–32]. Furthermore, a quasi-steady state approximation is used for the mass fractions of a few chemical species that are

too important to be neglected, but do not require advancement in space and time. The resultant reduced mechanism is hand-tuned to avoid non-linear iteration and also to obtain algebraic relations for the quasi-steady state species. Finally, on-the-fly stiffness removal in the DNS code [33,34] enables efficient explicit time integration, limited by the Courant–Friedrich–Levy (CFL) condition for chemical mechanisms with disparate timescales. Further computational cost savings may be obtained in the future through chemistry that is locally adaptive in time and space.

While for the most part, setting up a large-scale DNS is straightforward, there are some useful 'tricks-of-the-trade' that are commonly used. First, to avoid the issue of entrainment of fluid from outside of the computational domain through the physical boundaries, often mild grid stretching [11] is applied outside of the regions of physical interest, for example in a jet, to push the physical boundaries farther away and diminish their affect on the flame. Uniform grid spacing is imposed in the jet and flame regions for accuracy and for ease of postprocessing. Second, grid resolution is selected to provide adequate resolution for both the turbulence scales and the flame or ignition structure. Often the flame resolution is determined from one-dimensional strained flame calculations near steady strained extinction or ignition limits. Typically 0.5–2 grid points are required across the minimum Kolmogorov scales and at least five grid points are needed across the transported scalar with the largest gradient (often a minor species) in a flame. Even more stringent, the species reaction rates are resolved with at least five points across each consumption or production layer. Third, in setting up a three-dimensional production DNS run, coarse-mesh three-dimensional simulations at half-resolution are better approximations of the response of a turbulent flame than two-dimensional runs due to the intrinsic difference in the evolution of two-dimensional and three-dimensional turbulence.

Finally, there are multiple ways to generate turbulence feed data for a spatially-developing DNS run, e.g. a planar or round turbulent jet flame. One method is to first generate an auxiliary DNS of an inert turbulent channel or pipe flow to provide the reactive case with the initial turbulence field and inlet turbulence. Nominally, the inert flow requires less spatial resolution than the reactive DNS, and high-order interpolation is required as the feed data is sampled at the inlet of the reactive simulation. A short period of transition is usually required to establish the jet inlet plane velocity in the reactive DNS before initializing the flame. An alternative to feeding in realistic turbulence is to feed in velocity fluctuations obtained from a prescribed kinetic energy spectrum. Often this method is used in a shear flow to accelerate the development of shear generated turbulence. This method has the disadvantage of requiring a finite domain length for the prescribed spectrum to evolve into fully-developed turbulence, and therefore, one has to take great care in ensuring that regions containing artifacts of the turbulence initialization are excluded from statistical sampling. The most accurate method of feeding turbulence into a reactive flow DNS is to fully couple the inert flow solver with the reactive DNS solver so that there is full interaction between the acoustics, velocity and combustion. In [35], a spectral DNS solver for incompressible turbulence was coupled with a compressible reactive higher-order DNS code to perform DNS of a turbulent premixed V-flame developing in grid turbulence. In general, the treatment of complex geometry through multi-block DNS methods for reacting flows may be a method for coupling turbulence feed in confined geometries (e.g. pipe flow) to reactive flows in open domains (shear flows). In the multi-block approach, each sub-domain is discretized separately and appropriate coupling is achieved through interface conditions interior to the computational domain.

### *1.4. Low-Mach number DNS formulation with adaptive mesh refinement*

In combustion regimes, where the reaction fronts are much thinner than the finest turbulence scales, and more generally, where high gradient regions are restricted to a small volume of the computational domain, adaptive mesh refinement is computationally advantageous over a uniform grid. LMC is a massively parallel adaptive-mesh DNS code for combustion systems incorporating detailed chemical kinetics and transport in the low Mach number regime [36,37]. The low Mach number approximation exploits the natural separation of scales between fluid motion and the speed of sound, removing acoustics and the need to evolve them, from the analytic description of the system. Bulk compressibility effects due to chemical reaction and thermal conduction remain in the description but appear as a global constraint on the evolution of the velocity field. Assuming a mixture of perfect gases with a mixture-averaged model for differential species diffusion and ignoring Soret, Dufour, gravity and radiative transport processes, the low Mach number equations for an open domain are

$$\begin{aligned}
&\frac{\partial(\rho \boldsymbol{u})}{\partial t} + \nabla\cdot(\rho \boldsymbol{u}\boldsymbol{u}) = -\nabla p + \nabla\cdot\tau \\
&\frac{\partial(\rho Y_i)}{\partial t} + \nabla\cdot(\rho Y_i \boldsymbol{u}) = \nabla\cdot(\rho D_i \nabla Y_i) - \omega_i \\
&\frac{\partial(\rho h)}{\partial t} + \nabla\cdot(\rho h \boldsymbol{u}) = \nabla\cdot\left(\frac{\lambda}{c_p}\nabla h\right) \\
&\qquad\qquad + \sum_i \nabla\cdot\left[h_i\left(\rho D_i - \frac{\lambda}{c_p}\right)\nabla Y_i\right]
\end{aligned} \tag{1}$$

where $\rho$ is the density, $\boldsymbol{u}$ is the velocity, $Y_i$ is the mass fraction of species $i$, $h$ is the enthalpy of the gas mixture, and $\omega_i$ is the net destruction rate for species $i$ due to chemical reactions. Also, $\lambda$ is the thermal conductivity, $\tau$ is the stress tensor, $c_p$ is the specific heat of the mixture, and $h_i(T)$ and $D_i$ are the enthalpy and species mixture-averaged diffusion coefficients of species $i$ respectively where $T$ denotes temperature. These evolution equations are constrained by the ideal gas equation of state $p = \rho RT/W$ where $W$ is the mean molecular weight and $R$ is the universal gas constant.

The basic finite-volume numerical discretization [38] combines a symmetric operator-split treatment of chemistry and transport with a density-weighted approximate projection method to impose the evolution constraint. This integration scheme is embedded in an adaptive mesh refinement algorithm based on a hierarchical system of rectangular grid patches. The complete integration algorithm is second-order accurate in space and time, and discretely conserves species mass and enthalpy. The methodology has been implemented in parallel using a hybrid programming model. The data are distributed over nodes within the system using a coarse-grained load distribution strategy using MPI [39]. Within the nodes, fine-grained parallelization is accomplished using a thread-based approach implemented using OpenMP. This hybrid programming model enables the code to scale well to 50K processors.

## 2. Roles of DNS

In the following sections the various roles of DNS are illustrated through several examples related to: (1) stabilization of lifted turbulent jet flames in heated air coflow; (2) stabilization of a reactive transverse fuel jet-in-crossflow (JICF); (3) turbulent premixed combustion in the thin reaction zones regime and understanding NO formation; (4) stratified turbulent flame propagation; and (5) non-premixed turbulent combustion.

### *2.1. Stabilization of lifted turbulent jet flames in heated air coflow*

In many modern combustion systems such as diesel engines and gas turbines, fuel is injected into an environment of hot gases, and a flame may be stabilized through the recirculation of hot air and combustion products. Under such conditions, this leads to a turbulent lifted flame, and the hot environment admits the possibility of autoignition as a mechanism contributing to the stabilization of the flame base. Flame propagation coherent eddy structure have also been considered as possible mechanisms contributing to stabilization of the lifted flame base. Diverse fuel sources ranging from hydrogen to bio-fuels to traditional large hydrocarbons exhibit a wide range of ignition characteristics, and therefore can affect flame stabilization and lift-off heights in engines. Similarly, the amount of oxygen, dilution levels, and turbulence characteristics can also affect stabilization. In the following subsections recent highlights from DNS of turbulent hydrogen [13] and ethylene [40] lifted jet flames in heated coflow are described, a second-order conditional moment closure framework is used to analyze the source of turbulent fluctuations in composition and their effect on conditional reaction rates using the turbulent hydrogen DNS data [41], and an a posteriori evaluation of a multi-regime flamelet modeling approach for large-eddy simulation is presented [42] using the turbulent ethylene DNS data [40].

#### *2.1.1. DNS of turbulent ethylene lifted jet flame in heated air coflow*

Direct numerical simulation (DNS) of the near field of a three-dimensional spatially-developing turbulent ethylene [40] jet flame in highly-heated coflow was performed with a reduced mechanism to investigate the stabilization mechanism. The DNS was performed at a jet Reynolds number of 10,000 with over 1.29 billion grid points. Figure 1 shows instantaneous images of several scalar fields – scalar dissipation rate, mixture fraction, and hydroperoxy, methyl and hydroxyl intermediates – obtained from the DNS data. The results showed that auto-ignition in a fuel-lean mixture at the flame base is the main source of stabilization of the lifted jet flame.

This conclusion was verified by applying several diagnostics to the lifted jet flame DNS including the development and application of a chemical explosive mode (CEM) analysis [43,44], a Damköhler number defined by the CEM and the local scalar dissipation rate [43,44], and by Lagrangian tracking of key scalar and velocity statistics at the stabilization point [40].

A relatively new chemical diagnostic, chemical explosive mode analysis, was recently developed and applied to this flame [43,44]. The chemical explosive mode (CEM) is the eigenmode associated with a positive eigenvalue of the Jacobian of the chemical source term in a reactive flow. A reactive mixture in the presence of a CEM is intrinsically explosive and otherwise non-explosive, and therefore be classified as unburned and burned, respectively. CEM analysis (CEMA) was employed as a diagnostic to identify the spatial and temporal boundaries that separate burned and unburned mixtures in the turbulent jet flame. These boundaries unambiguously define ignition, extinction, and flame locations in complex turbulent flow fields.

A comparison of CEMA with conventional methods for identifying flame and ignition

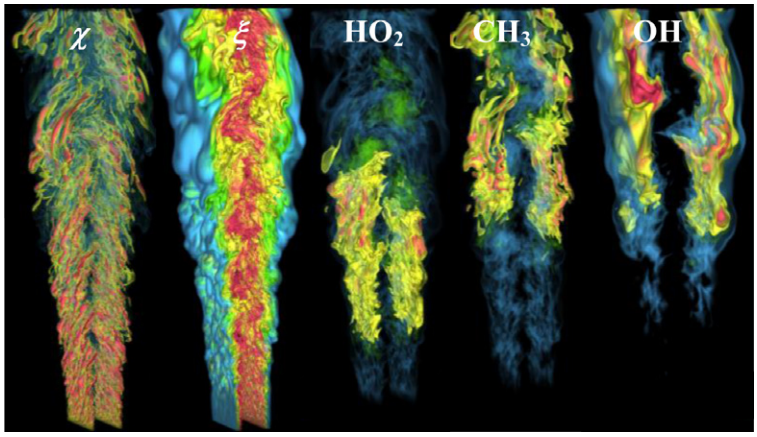


Fig. 1. Instantaneous volume rendering of scalar dissipation rate $\chi$, mixture fraction $\xi$, $HO_2$, $CH_3$ and OH radicals in a lifted ethylene/air jet flame (Re = 10,000) in a hot coflow [40].

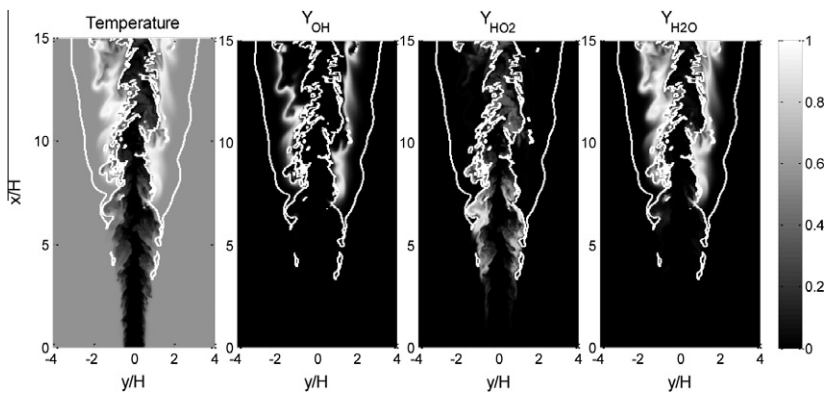


Fig. 2. Isocontours of normalized temperature and mass fraction of selected species in the lifted ethylene jet flame [44]. For each plotted quantity, the minimum and maximum values of the entire flow field are normalized to 0 and 1, respectively. The white isocontour indicates the premixed flames detected by CEMA.

structure is presented in Fig. 2, where the flame front detected by CEMA (white isoline) is superimposed onto the temperature, OH, $HO_2$ and $H_2O$ fields. It is seen that the lean flame front detected by CEMA, located well into the heated coflow shown in the gray background, is largely indiscernible from the temperature plot. This is because of the high coflow temperature (1550 K) and, consequently, the extremely lean flammability limit near the coflow. The heat release rate is negligible in the lean flame front and is undetectable by monitoring the temperature change. The high temperature zones, however, consist of the diffusion flame kernel, which involves near-stoichiometric mixtures. Therefore, CEMA can supplement traditional flame markers like temperature to provide more detailed description of flame strucuture.

It is also seen in Fig. 2(a, b and d) that CEMA can detect burned mixtures further upstream than locations where increases in temperature, OH and $H_2O$ concentrations are discernable, similar to observations made in DNS of a lifted hydrogen jet flame [13]. Nevertheless, in contrast to the hydrogen flame structure the ignited mixtures near the base of the ethylene jet flame show an intermittent topology, i.e. there are islands of ignited mixtures that are disconnected from the lifted flame base downstream. This topology suggests that the ignited mixture near the flame base is not first induced by upstream propagation of the flames downstream, but rather by autoignition in the upstream mixing layer. Finally, it is noted that while $HO_2$ is known to be important precursor for ignition, Fig. 2c shows that $HO_2$ peaks mainly in rich pre-ignition mixtures. As such it is not a good diagnostic for the detection of the lean fronts in the present ethylene jet flame.

In addition to auto-ignition, Lagrangian tracking of the flame base [40] revealed the passage of large-scale flow structures and their correlation with the fluctuations of the flame base similar to those observed in from DNS of a lifted hydrogen/air jet flame [13]. The ethylene lifted flame base exhibits a cyclic 'saw-tooth' shaped movement marked by rapid movement upstream and slower movement downstream. This is a consequence of the lifted flame being stabilized by a balance between consecutive auto-ignition events in hot fuel-lean mixtures and convection induced by the high speed jet and coflow velocities. This was confirmed by Lagrangian tracking of key variables including the flame-normal flow velocity ($-\mathbf{u}\cdot\mathbf{n}$), displacement speed ($S_d$), mixture fraction ($\xi$), and scalar dissipation rate ($\chi$) at the $z = 0$ plane presented in Fig. 3. Figure 3 shows that, in general, the stabilization point gradually migrates downstream and transversely outward following an abrupt movement upstream. This movement is associated with ignition of distinct kernels of mixture upstream of the existing flame base. Ignition is manifested by a large value of $S_d$ as shown in Fig. 3, dominated by the reaction rate contribution, $S_{Rd}$ (not shown). The stabilization point is also observed to move to a location of lower scalar dissipation rate and higher mixture fraction (although still fuel-lean) (not shown). The migration to higher mixture fraction is associated with the transverse, inward movement of the flame, and corresponds to a location with higher mean axial velocities. Once the flame base has been established at a new upstream location, and the displacement speed has decreased below the local fluid velocity, the kernel is convected downstream. While moving downstream the progress of reaction in the ignition kernel is retarded, either due to excessive dissipation rates or due to flame shortening resulting from entrainment or coherent jet mixing. The downstream movement of the stabilization point is primarily attributed to large local convective velocity, and intermittently, negative displacement speed associated with quenching from locally high mixing rates. In summary, the various DNS diagnostics suggest a conceptual picture of flame stabilization involving a cyclic pattern of rapid motion of the flame base upstream induced by an ignition event upstream of the flame base, followed by slower downstream movement of the flame base due to convection.

### 2.1.2. Three-dimensional DNS of a turbulent lifted hydrogen jet flame in heated coflow: flame stabilization and structure

By comparison with the ethylene lifted flame [40], a DNS of the near field of a three-dimensional spatially developing turbulent lifted hydrogen jet flame in heated coflow was performed with a detailed mechanism to determine the stabilization mechanism and the flame structure [13]. The DNS was performed at a jet Reynolds number of 11,000 with over 940 million grid points. The results show that auto-ignition, similar to the ethylene–air jet flame described above, in a fuel-lean mixture at the flame base is the main source of stabilization of the lifted jet flame at the high temperature, 1100 K, of the coflow stream. A chemical flux analysis showed the occurrence of near-isothermal chemical chain branching preceding thermal runaway upstream of the stabilization point, consistent with hydrogen auto-ignition occurring in the second limit. The Damköhler number and key intermediate-species behavior near the leading edge of the lifted flame also verified that auto-ignition occurs at the flame base. In addition to auto-ignition, Lagrangian tracking of the flame base revealed the passage of large-scale flow structures and their correlation with the fluctuations of the flame base. In particular, the relative position of the flame base and the coherent flow structure induces a cyclic motion of the flame base in the transverse and axial directions about a mean lift-off height. This was confirmed by Lagrangian tracking of key scalars, heat release rate and velocity at the stabilization point. Moreover, similar peaks in the Strouhal number were observed in comparisons of the power spectrum associated with the stabilization point fluctuation and the power spectrum of the correlation function oscillations between axial velocity fluctuations on

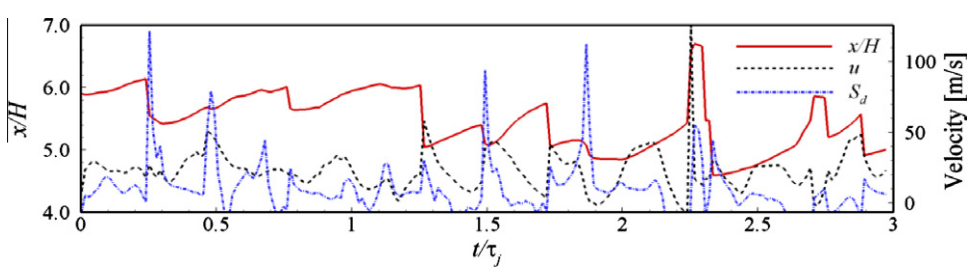


Fig. 3. Time history of the axial fluctuation in the flame stabilization height (x/$H$) where $H$ is the jet height, the local axial gas velocity (**u.n.**) and the displacement speed of the stabilization point ($s_d$) [40].

opposite sides of the jet at the mean stabilization height. The overlap of the dominant frequencies of both spectra suggest that the fluctuations of the stabilization point may be correlated with the passage of the large-scale flow structures near the lifted flame base.

While both the ethylene and hydrogen jet flames are stabilized by autoignition at the high coflow temperatures, the ethylene jet flame described above was unable to propagate upstream against the oncoming fast central jet and coflow streams, whereas the hydrogen jet flame was able to propagate upstream against the oncoming flow. This is attributed to the sensitivity of the stabilization mechanism to the coflow velocity and level of fuel jet dilution. The hydrogen jet flame had a comparatively lower coflow velocity against which the flame was able to propagate and less dilution of the fuel stream such that the stoichiometric mixture fraction was located farther away from the region of high mean shear.

#### *2.1.3. Analysis of second-order conditional moment closure applied to an autoignitive lifted hydrogen jet flame*

The timing and location of autoignition in the lifted hydrogen and ethylene turbulent jet flames described above is extremely sensitive to turbulent fluctuations of composition of key ignition intermediates and temperature. Second-order conditional moment closure (CMC) provides transport equations for conditional (co)variances in turbulent reacting flows and a useful framework to analyze the source and evolution of conditional scalar fluctuations and their effect on conditional reaction rates. CMC equations accounting for compressibility and differential diffusion were analyzed [41] using DNS data from the lifted turbulent hydrogen jet flame [13]. The analysis showed that at the flame base, second-order moments were required to accurately model the conditional reaction rates. Closure for the conditional reaction rates at first-order resulted in large error in the conditional reaction rates of all species and temperature. However, it was found that over 80% of the second-order reaction rate component was obtainable with a small subset (16%) of the species-temperature covariances. Reduction is important because the cost of including the entire conditional covariance matrix for all species is computationally intractable in engineering CFD simulations. An analysis of the balance of the second-order CMC equation further identified the initial source of generation of the conditional scalar variances [41]. In particular it was found that conditional spatial gradients for reacting and differentially diffusing scalars arise initially due to mean spatial variations in fluid history. For example, fluid mixture closer to the jet axis or further upstream encounters a shorter residence time, and hence, exhibits less reaction progress and diffusion. The conditional scalar fluctuations generated by mean variations in fluid history were subsequently amplified by reactions upstream of the stabilization location.

#### *2.1.4. A posteriori analysis of a multi-regime flamelet model for large-eddy simulation of lifted autoigniting jet flames*

Another promising modeling approach for treating mixed-mode combustion in lifted autoignitive jet flames is a multi-regime flamelet combustion subgrid model coupled with large eddy simulation (LES) (Knudsen TNF [42]). This approach is analyzed and assessed through comparison with the non-premixed $C_2H_4$ turbulent flame [40] described above. The LES of the lifted turbulent flame in a heated coflow tends to stabilize due to auto-ignition modulated by turbulent mixing.

Flamelet based models for LES that attempt to describe turbulence and auto-ignition interactions often make use of unsteady non-premixed flamelets. In statistically steady problems such as jet flames, this leads to modeling cost and complexity. Either the time scale from a single unsteady flamelet must be correlated with the distance from the jet, or a time-like variable must be used to parameterize a large, computationally expensive set of unsteady non-premixed solutions. Alternatively, these modeling issues can be circumvented by appropriately combining information from the pure steady non-premixed regime and from the pure auto-ignition regime. Separating these regimes eliminates the need to interpret the unsteady time scale, and dramatically reduces the cost of chemistry storage.

The LES case that is considered is the lifted $C_2H_4$ jet flame with a Reynolds number of $Re = 10{,}000$ discussed in Section 2.1.1 [40]. Results of the DNS demonstrated that, at the high temperature of the air coflow, auto-ignition is the primary flame stabilization mechanism. Additionally, however, results showed that turbulent mixing could locally inhibit the auto-ignition process, and lead to oscillations in the point. Several LES runs of this flame were performed on meshes consisting of between 1.4 million and 7.1 million cells. These runs used a combination of auto-ignition solutions and steady non-premixed flamelet solutions to describe reacting quantities such as the density and a progress variable source term. In general, solutions from these different regimes may be locally combined using a method proposed by Knudsen and Pitsch [45]. This method determines the combustion regime by explicitly considering which transport processes balance the chemical source term. Once known, information about the combustion regime can be used to map appropriate reacting quantities into the flow field. This method has the ability to quantitatively relate

the importance of auto-ignition and mixing. Two LES runs were performed to determine whether the limiting cases of this modeling approach envelop the DNS results. In these two runs, density was mapped into the LES flow field according to the local combustion regime, but the progress variable source term was taken only from pure auto-ignition solutions, or only from pure non-premixed flamelet solutions. A third LES run was performed that used combustion regime information to appropriately combine these source terms, and showed better agreement with the DNS data. Time averaged results from the three LES simulations were compared with the DNS results shown in Fig. 4. Figure 4 shows that all models adequately capture the velocity fluctuations, that the auto-ignition and steady non-premixed limits do envelop the DNS solution, and that both regimes are important. Using the combined model guided by the regime indicator provided the best agreement with the DNS data. For example, at locations close to the jet nozzle (small $X/H$), the LES based on the auto-ignition source terms over-predicts the conditional temperature and CO mass fraction in the DNS. This over-prediction occurs because the source term is not directly affected by dissipation and is relatively large. Conversely, the non-premixed source terms are affected by the dissipation rate in flamelet space. LES values based on these non-premixed source terms correctly predict temperature and CO near the nozzle, but under-predict it at downstream (large $X/H$) locations. This indicates that auto-ignition processes do occur in the flame, and must be accounted for if the lift-off height is to be predicted. The DNS data are being further mined to understand any discrepancies between conditional intermediate species.

### 2.2. Stabilization of a reactive transverse fuel jet-in-crossflow (JICF)

Another interesting configuration for studying lifted flame stabilization is a reactive transverse fuel jet in cross-flow (JICF). This was studied using three-dimensional direct numerical simulation (DNS) with detailed hydrogen chemical kinetics in order to investigate the mechanism of flame stabilization in the near field of a fuel jet nozzle [46]. JICF configurations are typically used in applications where high mixing rates are desirable between the jet and the crossflow fluids. Practical applications of JICF configurations are fuel injection nozzles and dilution holes in gas turbine combustors. Modern stationary gas turbine manufacturers are considering hydrogen-rich fuels which raises additional concerns about safety owing to the propensity of hydrogen to propagate upstream and stabilize above the fuel jet creating an undesirable situation. DNS [46] was performed of a nitrogen-diluted hydrogen transverse jet issuing from a nozzle into a cross-flow of heated air to understand the physical mechanism behind flame stabilization in the wake of a transverse fuel JICF. Analysis of the DNS revealed the interplay of flame structure and propagation characteristics with the flow and mixing fields in the near field of the fuel nozzle. JICF configurations are characterized by a high degree of intermittency and by large-scale coherent vortical structures that form in the wake of the transverse jet [47,48]. Through stretch, these flow structures result in an increase in the surface area of a scalar which leads to enhanced mixing rates, facilitating combustion. Previous DNS studies of inert JICF configurations [49] have shown that many parameters influence the flow and scalar mixing fields, e.g.: the

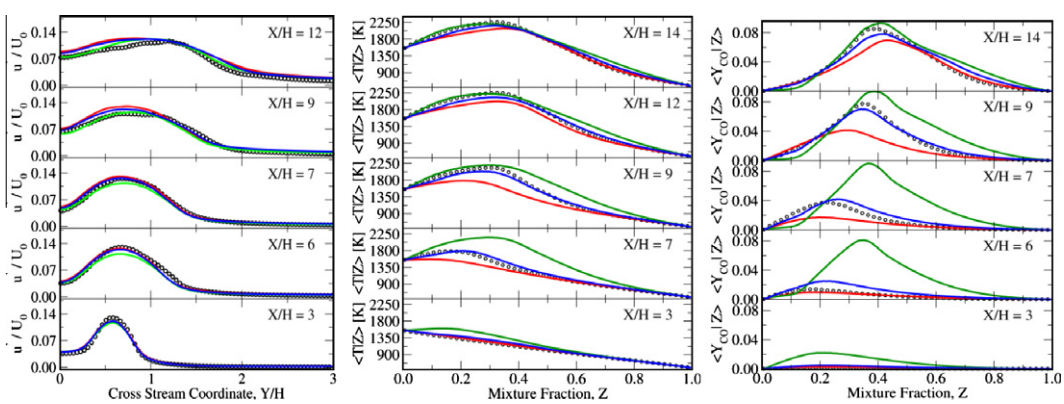


Fig. 4. Time averaged axial velocity fluctuations (left), conditional temperature (middle), and conditional CO mass fraction (right) in the flame at several axial position in the $C_2H_4$ jet flame [42]. DNS data (black symbols) is compared to LES simulations that use a progress variable source term from the pure auto-ignition limit (green line), from the pure steady non-premixed flamelets (red line), or from the combined auto-ignition and steady non-premixed flamelet regimes (blue line).

momentum ratio between the transverse jet and the cross flow, the shape of the nozzle, the injection angle, and the thickness of the approaching cross-flow boundary layer.

Direct numerical simulation (DNS) of the reactive JICF [46] provides detailed information about the complex flow structures and turbulence-chemistry interaction which facilitates flame stabilization. Figure 5 shows an instantaneous snapshot of the solution and combines volume renderings of the hydrogen, temperature and $HO_2$ radical scalar fields in a slice of the data through the midplane in the spanwise direction of the jet.

The resulting time averaged solution on the midplane in the spanwise direction is shown in the top (mixture fraction and normalized heat release rate) and bottom (normalized heat release rate and velocity magnitude) parts of Fig. 6. From the heat release rate isocontours, it is readily shown that the flame stabilizes, on average, at 1.5–2 jet widths downstream and 3–5 jet heights from the jet exit. The maximum heat release rate lies in a region where the magnitude of the average flow velocity is locally low and the average mixture fraction is near stoichiometric, $\xi = 0.171$ (the stoichiometric mixture fraction is $\xi_{st} = 0.169$). The isolines in Fig. 6 show where the magnitude of the mean velocity falls below 25 m/s, or approximately 10% of the maximum velocity in the domain. In the spanwise direction, the peak heat release is located on the center-line,

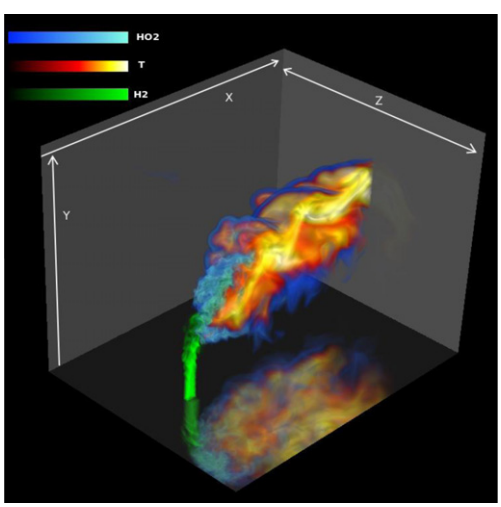


Fig. 5. Volume rendering of temperature (black body colormap), $HO_2$ (blue colormap), and $H_2$ (green colormap) scalar fields at $t = 2.802$ ms from start of simulation [46]. Opacity transfer functions adjusted to highlight the regions with high temperature, $HO_2$, or $H_2$ mass fraction.

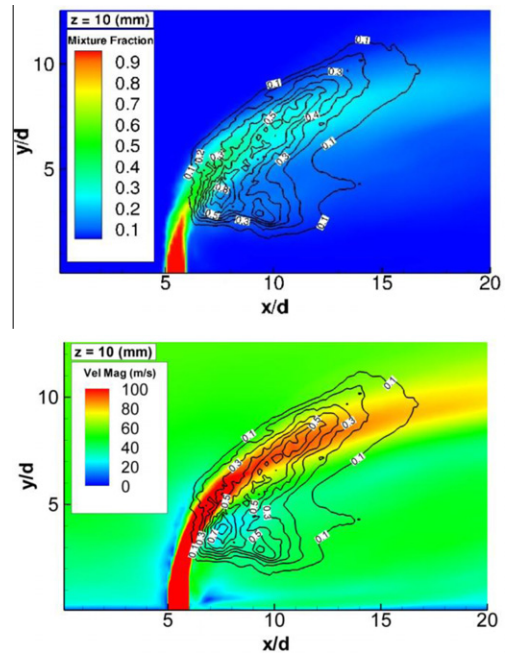


Fig. 6. Slices on the midplane showing Reynolds Averaged Navier–Stokes (RANS) solution for mixture fraction and normalized heat release rate (top) and velocity magnitude and normalized heat release rate (bottom) [46].

just above and between the large counter-rotating vortex pair which is known to enhance mixing between the jet and cross-flow fluids in nonreacting JICF.

The mean stabilization location suggests that the stabilization mechanism may be attributed to partially premixed flame propagation in a region where the convective velocity is low. The instantaneous solution is shown on the midplane in Fig. 7 where the Takeno flame index [50] (top) and mixture fraction scalar dissipation rate (bottom) are overlayed with contours of instantaneous heat release rate. The mixture fraction scalar dissipation rate is significant even in the far field, and especially where the heat release is large. At the same time, the Takeno flame index for this flame indicates that the gradients of the fuel and oxidizer are largely aligned in the reaction zone, i.e. where positive values close to unity indicate strong alignment (purple-red color). The alignment statistics taken together with the presence of thin regions with significant heat release where the fuel and oxidizer gradients are aligned substantiates that flame propagation is an important phenomena contributing to stabilization of this flame.

The flame stabilizes near the location where the jet breaks up and is strongly affected by the organized eddy structure, namely thin layers of the

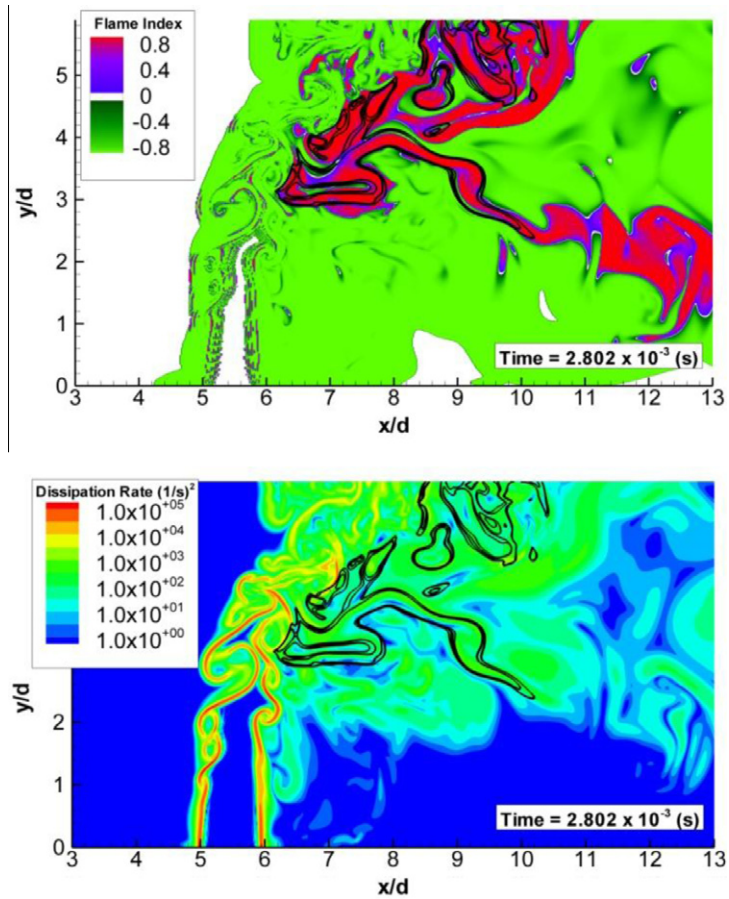


Fig. 7. Instantaneous snapshots of normalized flame index (top), scalar dissipation rate (bottom) [46]. Lines in top and bottom represent $8 \times 10^9$ and $1 \times 10^{10}$ J/m$^3$/s (approximately one-half of peak) isocontours of the instantaneous heat release rate.

flame are observed which are drawn into the vortical flow structures from the jet shear-layer instability. The strong interaction between the flow structure and the flame is verified through topological analysis of the velocity gradient tensor for this flow [46] and its spatial correlation with heat release rate.

### *2.3. Turbulent premixed combustion and NO formation in the thin reaction zones regime*

There are several interesting DNS configurations used to understand turbulent premixed combustion in the flamelet and thin reaction zones regimes. Laboratory-scale turbulent premixed flame simulations have been performed recently including a rod-stabilized V-flame [51], a piloted slot Bunsen burner [11,52] and the low-swirl burner [53,54]. In [55,56], thermodiffusively unstable flames were characterized across a broad range of forced turbulence and the manner in which the thermodiffusive instability affects the transition to distributed burning in an idealized configuration was described. Effects of Lewis number on the flame-turbulence interactions at high turbulence intensities were discussed in [56,57].

In [52] a low-Mach number adaptive mesh-refinement approach was used to simulate a turbulent methane–air slot burner in the corrugated flamelet regime and good comparisons were obtained with experimental measurements of the turbulent burning velocity, flame height and shape, and flame surface density with experiments by Filatyev et al. [58]. Sankaran et al. [11,59] performed DNS in a series of flames in a similar premixed, turbulent, spatially developing, slot-jet configuration in the thin reaction zones regime where intense turbulence can penetrate the flame's preheat layer [59]. The DNS are three-dimensional and also include a realistic reduced kinetic model of lean methane oxidation. These laboratory configurations offer two key advantages over the canonical freely propagating flames typically studied by DNS. First, the flame is statistically station-

ary, whereas freely propagating flames effectively see a different turbulence level depending on their location in the domain and are difficult to stabilize. Second, the mean shear sustains turbulence in the flame structure such that higher Karlovitz regimes are achievable. Results from Sankaran et al. [11,59] confirm that there is significant flame broadening in the preheat zone as a result of intense turbulence as revealed from conditional flame thickness statistics [11,59]. In the following subsections the role of DNS in premixed flame modeling is illustrated, showing how issues related to the flame surface density approach for LES in this intense turbulence regime are addressed [60,61], new relationships are derived to relate the statistical mean values of measured two-dimensional flame surface density function terms in the transport equation to those of the true three-dimensional quantity [62], a new model that relates intermediate species mixing frequencies to a progress variable mixing frequency required in the transported probability density function model is validated against the DNS data [63], and the generation of $NO_x$emissions from a lean hydrogen-air swirl-stabilized premixed flame [64] is analyzed.

### 2.3.1. Evaluation of LES models of turbulent premixed combustion in the thin reaction zones regime

One of the key issues in the modeling of turbulent premixed combustion is the accurate prediction of the unclosed reaction rate source term. The fluid-dynamic straining occurring in turbulent flows results in intense wrinkling of the flame, leading to large increases of the overall burning rate compared with the laminar situation. As discussed by others (e.g. [65–67]), in the situation when Kolmogorov scales are larger than the inner reaction layer thickness, chemical timescales can be assumed to be unmodified from their laminar values. Thus the closure problem can be reduced to the problem of predicting the flame area, cast as either a flame surface density [68] or a turbulent flame area ratio [65].

In the case of LES, part of the overall flame area is directly resolved on the computational grid, while the unresolved part must be modeled [69–71]. A key question then is what type of model needs to be adopted for the unresolved part. Putting aside differences in implementation, models can be divided conceptually into two kinds: those that assume that the sub-grid scales are in equilibrium, for example, turbulent flame speed based models [72–74], and those that consider non-equilibrium situations, for example transport equation models [75,76].

Using the Bunsen flame DNS described earlier [11,59], it is possible to directly assess the assumption of equilibrium, a question being addressed in [60]. The DNS results were filtered using a top-hat filter kernel and each term involved in the transport equation for flame surface density [70] was extracted. Inspection of these terms showed that except for a region close to the simulated burner exit, the transport equation was dominated by two large terms, namely, the effect of turbulent strain, and the combined action of curvature and flame propagation (curvature-stretch). As discussed in Ref. [77], these terms can be broken in different ways into resolved and sub-grid scale contributions. Here the sub-grid parts are given by:

$$a_{sg} = \overline{\left((\delta_{ij} - N_i N_j)\frac{\partial u_i}{\partial x_j}\right)_s \Sigma} - \left((\delta_{ij} - n_i n_j)\frac{\partial \tilde{u}_i}{\partial x_j}\right)\Sigma,$$

and

$$k_{sg} = \overline{\left(s_d \frac{\partial N_k}{\partial x_k}\right)_s \Sigma} - \overline{(s_d)_s}\frac{\partial n_k}{\partial x_k}\Sigma. \tag{3}$$

In Eq. (3), $\Sigma$ represents the flame surface density, $\underline{N} = -\nabla c/|\nabla c|$ represents the local flame normal vector, while $\underline{n} = -\nabla \bar{c}/|\nabla \bar{c}|$ is the normal to resolved progress variable isosurfaces. The local fluid velocity is $\underline{u}$ while the Favre filtered velocity is $\tilde{u}$. The displacement speed is given by $s_d$. The two quantities in Eq. (3) were locally extracted from the database, then Reynolds averaged by spatially averaging over the homogeneous spanwise direction exploiting symmetry about the center of the jet, and by averaging in time. Figure 8b shows these two quantities integrated across the transverse direction ($y$), for a range of different filter sizes, from 5dx to 21dx, where dx is the DNS grid spacing for case C in [59]. Also shown in Fig. 8a is the integral of the quantity $\Sigma - |\nabla \bar{c}|$, which represents the unresolved contribution to flame surface area. It may be observed that, as expected, the unresolved contributions to both flame surface density and the source terms become smaller with decreasing filter size. The second point to note is that the sub-grid strain and curvature-stretch terms nearly balance. There is a slight lag between the two such that the destruction of surface area occurs slightly further downstream than the production. Also notable is that as the filter size is decreased, the balance of strain and curvature deteriorates. However, from Fig. 8, it is observed that in this situation, the contribution of unresolved surface area is relatively small. As such for this particular flame, the assumption of equilibrium appears as a reasonable first approximation for the sub-grid flame surface area or flame wrinkling.

An important class of premixed combustion models that are based on an equilibrium assumption at sub-grid scales is those that attempt to model the sub-grid flame speed via an algebraic expression formed from quantities available at the resolved scales. There have been a number of

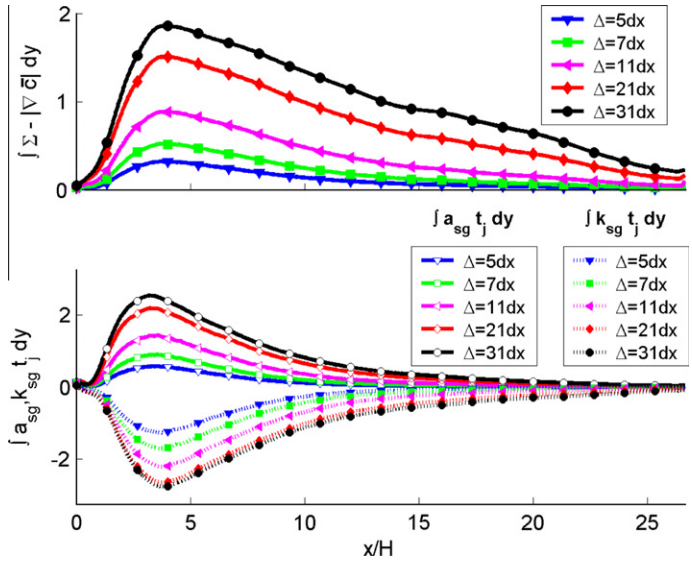


Fig. 8. Upper plate (a) $\int \Sigma - |\nabla \bar{c}| dy$ versus $x$. Lower plate (b) sub-grid contributions to the balance of area generation-strain term: $\int a_{sg} \Sigma dy$ and curvature-stretch term $\int k_{sg} \Sigma dy$ generated from Case C Bunsen DNS [61].

different proposals for this modeling, however they have been very rarely, if at all, considered in comparison to each other against the same data. Furthermore, there has been relatively little assessment of these models for high Karlovitz number flames in which the small turbulence scales can penetrate the preheat zone of the flame. Therefore, recent work has analyzed the Bunsen data-set to assess an array of different models in an *a priori* sense [61]. Four models are discussed here. The first model, which has been used extensively in thickened flame approaches is due to Colin et al. [78]:

$$\frac{s_T}{s_l^0} = 1 + \frac{2\ln(2)}{2C_{ms}(\mathrm{Re}_t^{1/2} - 1)} \Gamma\left(\frac{\Delta}{\delta_l}, \frac{u'_\Delta}{s_l^0}\right) \frac{u'_\Delta}{s_l^0} \tag{4}$$

where $s_T$ is the turbulent propagation rate, $\mathrm{Re}_t$ is the turbulent Reynolds number, $\Delta/\delta_l$ is the ratio of filter size to laminar flame thickness, $u'_{\Delta/s_l^0}$ is the ratio of the sub-grid velocity scale to laminar flame speed, $\Gamma$ is an efficiency function given in [73], and $C_{ms}$ is a constant. The second model is based on a fractal concept and is due to Charlette et al. [72,78]:

$$\frac{s_T}{s_l^0} = \Xi = \left(1 + \min\left(\frac{\Delta}{\delta_l^0}, \Gamma\left(\frac{\Delta}{\delta_l^0}, \frac{u'_\Delta}{s_l^0}\right) \frac{u'_\Delta}{s_l^0}\right)\right)^\beta, \tag{5}$$

where $\beta$ is related to the fractal dimension, $\Gamma$ is a (different) efficiency function, and $\delta_l^0$ is laminar flame thickness given in [72]. Pitsch [74] has championed the application of the level-set method, and has modeled the flame speed following concepts initially developed by Peters [65] for RANS simulations:

$$\frac{s_T}{s_l^0} = 1 - \frac{b_3^2 c_v c_s \Delta s_l^0}{2 b_1 Sc_t D} + \sqrt{\left(\frac{b_3^2 c_v c_s \Delta s_l^0}{2 b_1 Sc_t D}\right)^2 + \frac{b_3^2 \upsilon_t}{Sc_t D}}, \tag{6}$$

where $\upsilon_t$ is the sub-grid turbulent viscosity, $Sc_t$ is the turbulent Schmidt number, $D$ is the diffusivity, $c_v c_s$ is a Smagorinsky-type constant. The constants $b_1$ and $b_3$ were originally determined from expectations based on RANS, which over-predicted the flame speed when compared with this data-set, so the parameters were adjusted to have values 1.0 and 0.15, respectively. Finally, a new model developed by Hawkes [61] was assessed. This model uses a power-law type approach similar to Charlette's model, extending this concept to the thin reaction zones regime. In the high Ka limit, where the present data-set are located, this model reduces to:

$$\frac{s_T}{s_l^0} = \Xi = \left(1 + c\frac{\Delta}{\eta_i}\right)^\beta, \quad \text{where}$$

$$\eta_i = \left(\frac{D^3}{\varepsilon}\right)^{1/4} \tag{7}$$

is the Obukhov–Corrsin scale [65] and $\beta = 2/3$. These choices correspond to the situation where the flame behaves essentially as a passive scalar, and it may be shown that at high levels of wrinkling, the limiting flame speed is the same as that proposed by Pitsch [74], i.e. Damköhler's small-scale limit.

These four models were evaluated against the DNS in an a-priori test for a range of different fil-

ter sizes, from 5dx to 21dx, for case C. Figure 9 shows the results, which have been averaged in the same way as Fig. 8.

Up to a filter size of 11dx all four models show encouraging results. This filter size corresponds best to what is normally termed "LES" since it resolves approximately 80% of the kinetic energy. In part this good agreement is due to the fact that, as seen in Fig. 8a, a large fraction of the flame surface area is resolved. The agreement begins to deteriorate at larger filter sizes, with the models due to Hawkes and Pitsch, which were both developed with intent to apply in this regime, offering slightly better agreement. Ongoing work is assessing these and several other models against the same data, as well as using a Germano-type dynamic procedure to determine one of the constants [79].

*2.3.2. Estimates of the three-dimensional flame surface density and all terms in its transport equation from two-dimensional measurements*

Given its importance in premixed flame models, there has been considerable interest in the experimental measurement of the flame surface density and key quantities in its transport equation, namely strain rate, curvature, and displacement speed. Because of the difficulty involved in obtaining three-dimensional measurements, the vast majority of this work has been performed in two spatial dimensions. Therefore a mapping that relates the two-dimensional measurement to the three-dimensional reality is needed. Recently, new relationships were developed to relate the statistical mean values of measured two-dimensional quantities to those of the true three-dimensional quantity in the case of isotropic scalar fields and turbulence [62]. The relationships were derived for the flame surface density as well as for the terms in its transport equation. DNS of turbulent premixed methane–air slot-jet Bunsen flames were employed [11,59] to assess the performance of the relationships. In all cases considered, the agreement was found to be good and improved with downstream distance, $x/H$, and with jet Reynolds number as shown in Fig. 10.

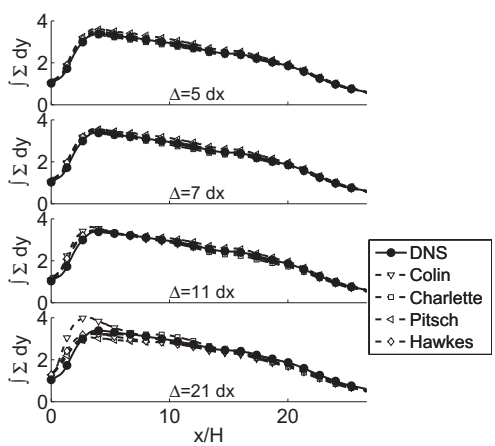


Fig. 9. A priori comparison of DNS and models for the turbulent flame speed in LES [61].

*2.3.3. A priori analysis of reaction–diffusion effects on species mixing frequency in turbulent premixed methane–air flames*

Another central modeling quantity is the scalar mixing frequency that is required, for example, in transported probability density function calculations applied to turbulent premixed combustion. Three-dimensional turbulent Bunsen flame DNS data with reduced methane-air chemistry [11,59] in the thin reaction zones regime were analyzed to assess previous findings regarding the progress variable dissipation rate [63]. The dissipation rate was found to be controlled by a balance between molecular dissipation and gradient generation due to flame propagation, dilatation and compression of scalar gradients by turbulent strain, consistent with previous findings from two-dimensional and simple chemistry DNS analyses.

However, the mixing rates of intermediate species were found to be significantly larger, up to a factor of ten greater than those of progress variable. They are not well modeled by existing models for mixing rates of either the progress variable or of passive scalars. For premixed combustion in the thin reaction zones regime where the turbulence is intense dilatation effects are negligible for the intermediate species. The effect of turbulent straining is also less important, becoming negligible for high Damköhler number species. Instead, a reaction-dissipation balance controls

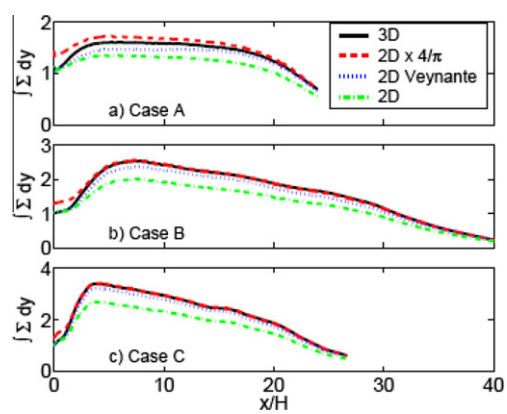


Fig. 10. Comparison of the Reynolds averaged 3D flame surface density and its 2D estimates [62] (red – present isotropic correction; blue – correction by Veynante [80]) for Case A ($Re_j = 840$), Case B ($Re_j = 1400$) and Case C ($Re_j = 2100$) from DNS of a lean methane-air slot Bunsen flame [11,59]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the intermediate scalar gradients, governed by the intrinsic premixed flame structure.

The DNS analysis of the mixing rates of intermediate species led to the development of a model that relates their mixing rates to that of a progress variable [63]. The model uses the species gradients obtained from laminar flames to estimate the relative magnitude of the species dissipation rates in the turbulent flame. The implied alignment of the species and progress variable was assessed against the DNS data and found to provide a reasonable approximation since scalar gradients parallel to the flame have negligible contribution to the average dissipation rate. The use of laminar flame data provides a good approximation for the relative magnitude of the species gradients, even in the thin reaction zones regime. The DNS analysis showed that the new model accurately captures the variation of the intermediate progress variable mixing frequencies for premixed flames with sufficiently high Damköhler numbers.

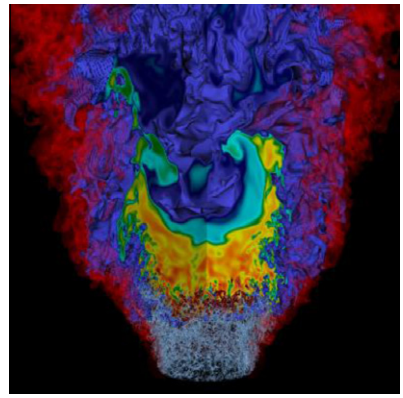

Fig. 11. Cutaway of hydrogen flame snapshot [64]. Isovolume (enclosed in purple, colored on exposed surfaces) of NO concentration, where the fuel boundary is the semi-opaque red surface, and grey represents vortical structures in the shear layer of the turbulent swirling flow.

*2.3.4. Lean turbulent hydrogen premixed combustion and NO generation*

Bell and Day [64] have recently performed a simulation of a low-swirl burner fueled by hydrogen at an equivalence ratio of 0.37. A detailed chemical description containing 15 species and 58 reversible Arrhenius reactions was used to describe premixed hydrogen–air combustion and subsequent $NO_x$ formation. This mechanism was derived by rate-of-production and sensitivity analyses from a full reaction mechanism [81,82] to obtain the minimum reaction set that describes NO formation in $H_2$ oxidation at lean conditions. It includes NO formation routes through NNH and $N_2O$, as well as the thermal mechanism. Details of the mechanism along with two dimensional studies of $NO_x$ emissions in $H_2$ flame using this mechanism were presented in [83].

The simulation domain measures 25 cm × 250 cm × 25 cm and is discretized using a base mesh of $256^3$. Three levels of grid refinement, each by a factor of two, dynamically track regions of high vorticity (turbulence) with an additional fourth level added to capture the reaction zone, resulting in an effective resolution is $4096^3$. Note that this configuration is identical to the high-turbulence lean hydrogen case discussed in [54] except for the inclusion of detailed emission chemistry. The overall flame shape, and structure of the NO profile is shown in Fig. 11.

To analyze the details of the nitrogen chemistry, the procedure outlined in [84] was used to partition the reaction zone above the nozzle ($r < 2.5$cm) into "flamelets" – small volumes that are bounded by integral curves along the temperature gradient. The construction presumes that temperature is a suitable progress measure. Along this progress coordinate, the temperature and local fuel consumption are sampled, in addition to the progress rates for all reactions in the model that involve nitrogen chemistry. The flamelet data is sorted and binned according to the peak fuel consumption rate in each flamelet, and conditionally averaged in temperature space. These peaks were found to be as much as a factor of 13 higher than the corresponding flat laminar flame and correspond to areas of intense burning characteristic of the cellular structure of thermodiffusively unstable flames.

Not surprisingly, the formation of $NO_x$ emissions increased with the local burning enhancement. To quantify the overall $NO_x$ production, Fig. 12(a) shows the integrated NO production as a function of temperature, showing that the majority of NO is generated near 1500 K. Also, approximately 80% of the NO is created through the NNH route, while 20% is generated via the thermal route.

A more quantitative description of how the emission characteristics vary with burning intensity is obtained by separating the flamelet into three bins corresponding to weak, medium and strong burning and computing integrated reaction path diagrams over each region, which are shown in Fig. 12(b). Here the thickness of the graph edges is scaled to the transfer rate of nitrogen atoms between species pairs. The data are integrated over the flame region where $r < 2.5$ cm, and over the entire temperature range. Contributions to each edge are collected into three burning groups (see caption), and indicated in percentages on the figure. The $N_2 \rightarrow N_2O$ path shows a negative contribution for the weakly burning region, indicating that the path actually reverses. The results show that in spite of the extremely large

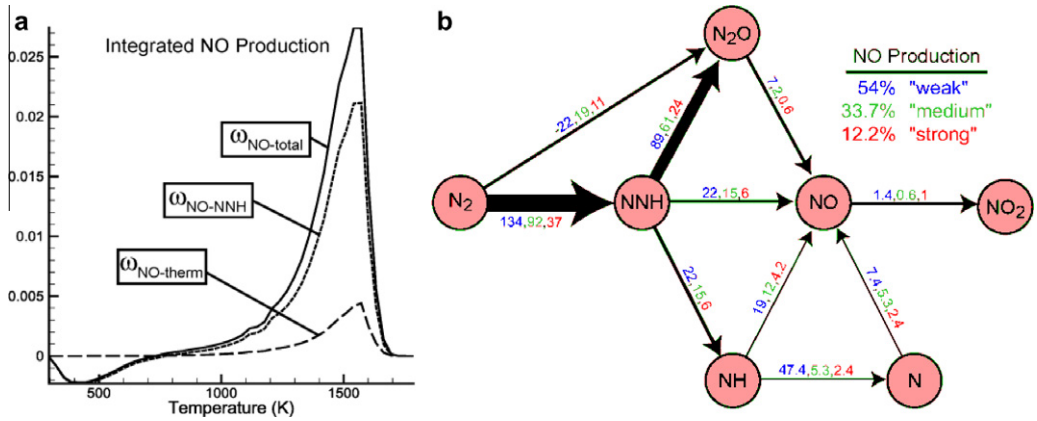


Fig. 12. (a) Distribution of integrated emissions production and (b) Path diagram quantifying the integrated transfer rate of N atoms between species, normalized by the total rate of N atoms into NO [64]. Each path indicates the contribution (in percent of NO production rate) from the weak ($\omega_{Peak} = 0$–$4\omega_{Peak,0.37}$, blue), intermediate ($\omega_{Peak} = 4$–$9\omega_{Peak,0.37}$, green) and strong ($\omega_{Peak} = 9$ –$13\omega_{Peak,0.37}$, red) flames, where $\omega_{Peak,0.37}$ is the peak consumption rate of the flat flame at $\Phi = 0.37$.

NO production rates in the hotspot regions, the weakly burning parts of the flame account for over half of the total NO production, because the strongly burning segments only represent a small portion of the flame volume.

### 2.4. Stratified combustion

In direct injection engines, fuel is injected directly into the combustion chamber to generate a highly stratified fuel-air mixture prior to ignition. This can give more reliable ignition than alternative dilute, homogeneous combustion modes. The reduced fuel consumption, particularly at low-speed light-load operation, is of great interest to both spark-ignition gasoline and compression ignition diesel engine manufacturers. In stratified combustion, the fuel and oxidizer are neither completely mixed nor completely segregated prior to combustion, and therefore traditional modeling approaches based on purely 'premixed' or 'nonpremixed' burning modes are inadequate to describe it. Another example of stratified combustion is found in lean partially premixed gas turbines used for power generation where locally richer mixtures are used to increase flame stability, while the globally lean flame gives extremely low $NO_x$ emissions.

Present combustion devices have been highly optimized for the range of fossil fuels currently available. Use of new fuels such as hydrogen, and synthetic or bio-fuels requires different condi-

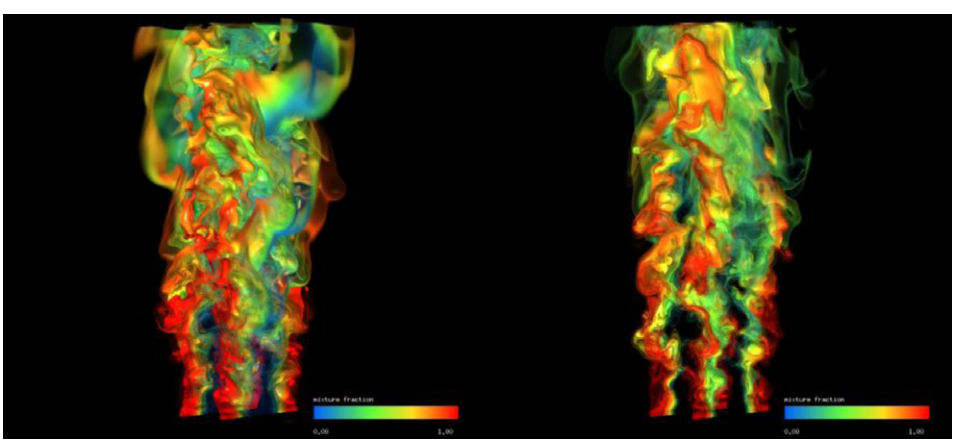

Fig. 13. Volume rendering of the heat release field colored by equivalence ratio for the low stratification case on the left (blue = 0.41, red = 1.0), and the high stratification case on the right (blue = 0.0, red = 1.4) [85].

tions for effective and clean burning. Fuel stratification is being investigated as a potential strategy for meeting the combined constraints of anticipated fuels, pollutant emission, and fuel consumption, but models for stratified combustion are still lacking, and require considerable development and validation. For example, fundamental knowledge regarding the influence of both variations in equivalence ratio and in its gradient on flame structure, burning rates, and propagation speeds are needed, as recently obtained from DNS of a turbulent stratified methane-air slot jet flame and from one-dimensional strained laminar stratified flames in a counterflow configuration.

DNS of a turbulent stratified methane–air slot jet flame [85] was performed with a reduced methane–air mechanism to provide understanding of the stratified combustion regime. The benchmark simulation data resulting from this study provides a unique test-bed for predictive engineering models in the challenging partially-premixed regime.

Figure 13 shows the simulated stratified flame surface colored by equivalence ratio $\phi$, which indicates the varying fuel-air concentration. The configuration involves a turbulent slot Bunsen fuel jet issuing into a co-flow of burnt products. This configuration allows for intense turbulence to interact with the flame front while preventing the flame from blowing out. Stratification was achieved by varying the equivalence ratio along the spanwise ($z$) direction of the slot. The stratification levels studied were referenced to the uniformly premixed $\phi = 0.7$ case investigated in [11,59], including a low stratification case $0.4 < \phi < 1.0$, and a highly stratified case with equivalence ratio ranging from zero (pure air) to 1.4. Highly efficient, reduced methane-air chemical kinetic models were optimized and validated for the simulation conditions [85]. These models include 13 or 19 transported species, with the 19 species mechanism being used to model rich flames including their production of nitrogen oxides.

The jet slot width of 1.8 mm and a mean jet velocity of 100 m/s resulted in a moderate jet Reynolds number of 2100. The flame was highly wrinkled with significant flame–flame interactions [11,59]. Under these conditions the flame is characterized by the thin reaction zones regime in which turbulent eddies penetrate the flame's preheat layer but are too large to disrupt the reaction zone. This resulted in a thickened preheat layer identified by the reduction of the conditionally averaged progress variable gradient, compared to the laminar flame gradient. As the equivalence ratio fell to $\phi = 0.4$ and subsequently dropped below the lean flammability limit there was a dramatic reduction in the heat release rate. The areas of weakly burning flame were more susceptible to disruption by the turbulent motions, permitting study of the broken reaction zones combustion regime.

As the initial fuel–air stratification dissipated due to turbulent mixing, the flames were subjected to steep, time varying equivalence ratio gradients. Locations where the equivalence ratio gradients align normal to the flame exhibited greatly modified burning rates and flame propagation speeds, $S_d$ compared to the perfectly premixed case. Figure 14 shows the variation of propagation speed with the flame normal equivalence ratio gradient, indicating enhanced flame speeds when the products are richer than the reactants, with the reverse also true. The figure's symbols are colored by the strain rate normal to the flame since the magnitude of the equivalence ratio gradient, but not its orientation, is strongly correlated with the local fluid dynamic strain. Studying this effect using computations of strained one-dimensional counterflow flames in the presence of equivalence ratio gradients [86] revealed the thermo-chemical basis for this observation. Specifically, equivalence ratio gradients resulting in flames with richer, higher temperature products lead to enhanced equilibrium concentrations of species such as the hydroxyl radical in the flame's carbon monoxide–hydrogen recombination layer. Diffusion of these highly reactive intermediate species towards the reactant side of the flame, as opposed to thermal diffusion, is the main source of the increased reaction rate. Data from the one-dimensional simulations are included in Fig. 14. The good agreement with the DNS data shows that the stratified counterflow configuration is a valid paradigm for understanding and modeling turbulent combustion with fuel–air stratification.

### 2.5. Non-premixed tubulent combustion

In this section DNS of turbulent non-premixed combustion, where the fuel and oxidizer streams are segregated, is presented, focusing on finite-rate

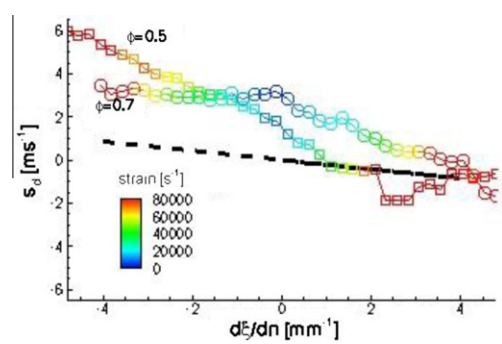


Fig. 14. Variation of displacement speed $S_d$ with the flame normal equivalence ratio gradient [85]. Data are conditionally averaged on the 1400 K iso-surface and $\phi = 0.7$ in the low stratification case (circles). Symbols are colored by the flame normal strain rate. Data for one-dimensional counterflow computations [86] are plotted with a black dashed line.

chemical interaction with turbulent mixing. DNS of extinction and reignition phenomena in turbulent jet flames [10,87–90] are first presented, followed by an analysis of reactive mixing timescale in turbulent jet flames [10], an assessment of an artificial neural network (ANN) chemical tabulation scheme used as a linear-eddy subgrid model for a large-eddy simulation of turbulent non-premixed jet flames with strong finite-rate kinetic effects [91], an a priori DNS evaluation of a one-dimensional turbulence model that captures extinction/reignition processes in jet flames [92], and a method to extract 3D scalar dissipation rate statistics from lower dimensional measurements validated against DNS of extinction/reignition jet flames and 2D measurements [93].

*2.5.1. Extinction and reignition in turbulent jet flames*

Unlike premixed flames efficient mixing is critical in non-premixed flames where the fuel and oxidizer streams are segregated and molecular mixing is required for chemical reaction. High molecular mixing rates characteristic of turbulent flows enhance reactions rates, and thereby improve combustion efficiency. However, the interaction between finite-rate chemistry and excessive mixing rates can culminate in local extinction, and if pervasive, can lead to global blowout, destabilization of the flame, and increased generation of emissions. Understanding the coupling between chemistry and fine-scale turbulence and mixing has been a focal point of measurements of the scalar dissipation rate, mixture fraction, and flame markers [94–99]. However, state-of-the-art measurements of passive and reactive scalar mixing are still limited to lower dimensionality, and time-resolved measurements of the intermittent processes of extinction and reignition are still largely limited to 10 kHz high-repetition imaging of cross-planar measurements along a line or in plane using 2D laser induced fluorescence measurements [100,101]. Simultaneous velocity and scalar measurements are even more scarce and limited to two spatial dimensions. Therefore, there is a unique role for 3D time-resolved DNS with complex chemistry to investigate finite-rate kinetics interactions with turbulent mixing in jet diffusion flames.

Past DNS [10,87–90] studies have investigated extinction/reignition phenomena in turbulent nonpremixed flames. Sripakagorn et al. [90] attributed extinction to local strong fluctuations in the scalar dissipation rate. Subsequently, Pantano [89] performed DNS of a spatially developing methane–air jet flame with a reduced 8 species, 4-step reduced mechanism and observed that the flame edge speed is negatively correlated with the scalar dissipation rate, and that local extinction holes reignite primarily by edge flame propagation tangential to the flame surface. Recently, massively parallel DNS of turbulent nonpremixed plane temporally-evolving jet flames were performed with skeletal syngas kinetics with 11 species and 21 reaction steps [10,87,88]. Up to 500 million grid points were employed, utilizing up to 4096 computer processors, and jet flames with Reynolds numbers of up to 9000 were computed with good resolution. The simulations feature strong finite-rate chemistry effects including extinction and reignition, and a parametric study with increasing Reynolds number, holding Damköhler number fixed, was performed.

The DNS data were used to understand reactive scalar mixing in turbulent jet flames and to investigate the distinct modes of reignition following local extinction. The total extinguished flame area based upon the threshold value of a reactive scalar, the hydroxyl radical, on an iso-mixture fraction surface, was the metric used to determine the total amount of extinction. This metric was also used to understand the motion of the flame edges that form the boundaries between the burning and nonburning regions. Statistical analysis of the joint probability density function of the scalar dissipation rate and the edge flame speed showed a transition in behavior not previously observed. Early on after the formation of extinction holes a strong negative correlation developed between scalar dissipation rate and edge flame speed, consistent with spatially-developing DNS of a methane-air jet flame undergoing extinction and reignition [89] as shown in Fig. 15 (left image). Later in the reignition process the predominant correlation between scalar dissipation rate, for low to moderate values, and flame edge speed became positive, with only the highest dissipation rates exhibiting a negative correlation with edge speed (see Fig. 15 (right image)). Examination of alignment between the normal vectors of the iso-mixture fraction surface and the reacting scalar isosurface confirmed that the source of the positive correlation is due to turbulent compressive straining folding burning surfaces onto non-burning mixture surfaces, and hence, providing sources of heat and radicals normal to the flame surface. This reignition mechanism is similar to the 'flame engulfment' mode observed in simple chemistry DNS by Sripakagorn [90]. Reignition also occur red, albeit to a lesser extent since the turbulence intensity was much greater than the laminar burning velocity, at low scalar dissipation rate through propagation of flame edges tangential to the flame surface [88].

In a recent related numerical study of a model one-dimensional configuration used to explore the reignition of a cold reactant mixture adjacent to a burning non-premixed flame with one-and two-step global step kinetics, a simple scaling estimate was put forth by Liao and Pantano [102] describing the relative importance of the flame folding, lateral re-ignition mode in comparison with the

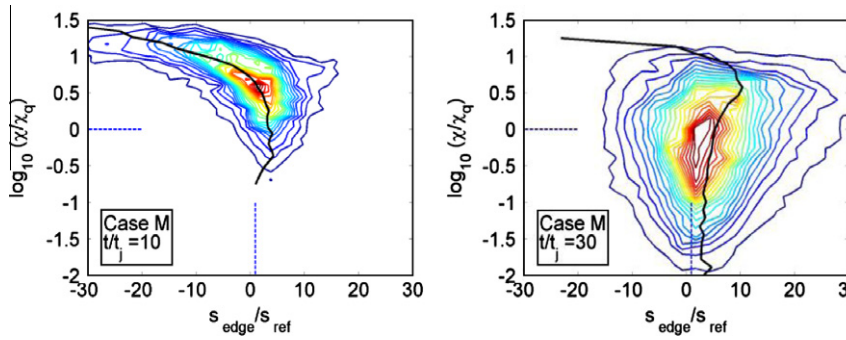


Fig. 15. Color contours: the joint pdf of scalar dissipation rate and edge flame speed for two simulation times, from left to right, (lots of extinction and early reignition) 10 jet times and (late reignition) 30 jet times [87,88].

edge flame propagation mode. Their suggested scaling implies that for extinction holes larger than the Taylor microscale the former recovery mechanism is more likely, provided that turbulent advection brings together the stoichiometric burning flame surface in close proximity with the cold reactant mixture, and conversely, the latter is more probable for small holes.

*2.5.2. Mixing time scales of reactive scalars in turbulent non-premixed flames*

Similar to the analysis of mixing time scales of reactive scalars in a turbulent premixed flame [63], the mechanical-to-scalar mixing timescale ratio, a central quantity used in the transported probability density function (pdf) approach [103], was evaluated for conserved as well as reactive scalars in turbulent non-premixed flames using the $CO/H_2$ planar jet flame DNS database [10]. The influences of molecular diffusivity and reaction were investigated. Nominally, the mixing time scale is assumed to be identical for all scalars, and the same order of magnitude as the large scale turbulence timescale. However, in flames, differential diffusion and the strong interplay of mixing and reaction may invalidate these assumptions. It is difficult to directly assess these assumptions in a posteriori tests, and measurements of reactive scalar mixing are not yet possible for all species. Therefore, DNS of jet flames undergoing extinction and reignition were used to evaluate such assumptions.

The mechanical-to-scalar mixing time-scale ratio for a conserved scalar (mixture fraction) was found to be order unity and nearly independent of Reynolds number (Re) at sufficiently high Re. On the other hand, nonunity Schmidt numbers and chemical reactions were found to affect the mechanical-to-scalar mixing timescale ratios of reactive species. For major species, the dependence of the timescale ratio on molecular diffusivity declined with increasing Reynolds number, Re, although a weak dependence persisted for the highest Re simulated (Re = 9000). Chemical kinetic effects on reactive scalar mixing were found to be most prominent for key radical species, however, and these persisted to high Re as they involveed strong interplay between diffusion and reaction. The mixing attributes of these species were found to depend not only on a balance between local turbulence production and molecular dissipation, but also on reaction. These results point to the importance of considering the coupling between diffusion and reaction, particularly in the presence of strong finite-rate chemistry effects.

*2.5.3. Large eddy simulation of extinction and reignition with artificial neural networks based chemical kinetics*

Large eddy simulation (LES) of a non-premixed, temporally evolving, planar $CO/H_2$ flame was performed to evaluate the sub-grid chemistry computations using an artificial neural networks (ANN) approach [91]. An a posteriori comparison of the LES with a direct numerical simulation (DNS) study [10,87,88] was performed. The DNS exhibited strong finite-rate kinetic interaction with turbulence resulting in significant amounts of local extinction and reignition, and hence, provided a stringent test case. The chemical kinetics modeling with ANN was based on a recent approach [91], and replaces a stiff ODE solver to predict the species reaction rates in the subgrid linear eddy mixing (LEM) model based LES (LEM-LES). Conditional statistics of some of the key species and temperature were extracted from the DNS data [10] and compared with the LEM-LES using ANN. The results show that chemistry tabulation with ANN captures the correct extinction and reignition physics with good accuracy compared to the DNS, and provides a significant speed-up compared with direct integration methods.

*2.5.4. An evaluation of the one-dimensional turbulence model: comparison with direct numerical simulations of $CO/H_2$ jets with extinction and reignition*

A variant of the One-Dimensional Turbulence (ODT) model formulated in an Eulerian reference

frame was applied to a planar nonpremixed turbulent jet flame and results from the model prediction [92] were also compared with recent $CO/H_2$ temporal slot jet flame DNS undergoing extinction and reignition [10,87,88]. The ODT model solves the full set of conservation equations for mass, momentum, energy, and species along a one-dimensional domain corresponding to the transverse jet direction. The effects of turbulent mixing were modeled via a stochastic process, while the full range of diffusive-reactive length and time scales were resolved directly on the one-dimensional domain. The ODT model used the same detailed chemical mechanism and mixture averaged transport used in the DNS [10]. Comparisons between the model and DNS data in physical and mixture fraction space were made, including conditional statistics [92]. Results verify that the model accurately reproduces the main features of DNS dataset, and captures the trends for extinction and re-ignition as exhibited in the evolution of the conditional mean temperature and radical profiles. The primary difference observed between the model prediction and DNS data were attributed to the early excess extinction observed in the model. It was hypothesized that the difference may be due to the inability of the one-dimensional ODT model to capture inherently multi-dimensional vortex pairing processes resulting from the shear instability.

#### *2.5.5. An analysis of lower-dimensional approximations to the scalar dissipation rate using DNS of plane jet flames*

A central quantity in turbulent non-premixed combustion is the scalar dissipation rate, a measure of the local mixing rate, competitive with reaction rate. The Damköhler number, the ratio of the reaction rate to dissipation rate, determines the burning intensity and below a critical threshold, extinction occurs in a flame. Experimental measurements of the scalar dissipation rate in turbulent flames are limited to estimates of the true three-dimensional (3D) scalar dissipation rate from lower-dimensional gradient measurements, either in one-dimension (1D) or in two-dimensions [94–99]. New relationships were developed to relate the true values and their lower dimensional approximations by assuming a form for the statistics of the gradient vector orientation, which enables several new results to be obtained and the true 3D scalar dissipation PDF to be reconstructed from the lower-dimensional approximations [93]. Direct numerical simulations (DNS) of turbulent plane jet flames at $Re = 9000$ [10] were used to generate the orientation statistics, and to assess assumptions and the reconstructed results. New theoretical relationships between the lower-dimensional and true moments of the scalar dissipation PDF assuming a log-normal true PDF were developed and tested [93]. PDFs reconstructed from lower-dimensional gradient projections were compared with the true values and excellent agreement was obtained for both 1D and 2D simulated measurements perpendicular to the mean flow variations. Comparisons of PDFs of thermal dissipation from DNS with those obtained via reconstruction from 2D experimental measurements [98,99] showed good agreement, indicating this PDF is more generally applicable to different flow configurations. These comparisons are shown in Fig. 16 along with isocontours of the logarithm of the normalized scalar dissipation rate in a spanwise, $z$, and streamwise, $x$, plane. A new theoretical relationship was also developed to reconstruct the joint PDF of the scalar dissipation and any other scalar, for instance, any chemical species or temperature. This was assessed by reconstructing the conditional means of the hydroxyl mass fraction and comparing it with the actual value, and excellent agreement was obtained.

## 3. Future outlook for DNS

### *3.1. Challenges of exascale computing and implications for DNS*

Pushing the DNS envelope to higher Reynolds numbers, higher pressures, and greater chemical complexity will require significant increases in computing power over current petascale supercomputers. Elucidating fundamental 'turbulence-chemistry' interactions in combustion relevant to the development of a predictive modeling capability for the design of fuel efficient, clean burning combustion devices utilizing diverse fuels will require at least a thousandfold increase in computing power over the next decade, that is, exascale ($10^{18}$ floating point operations per second) computing. Achieving exascale computing power cannot be accomplished with incremental improvements to current hardware and software. With clock rates leveling off at several gigahertz, systems will require more than one billion concurrent operations to achieve exascale levels of performance. Most of this increase in concurrency will be within the compute node, hence requiring applications to increasingly exploit intra-node parallelism. Other critical issues include reducing power consumption by roughly a factor of 300 (petascale computers consume roughly 7 MW whereas exascale computers need to meet a practical power limit of 20 MW), resiliency to faults or data corruption (roughly a factor of 100-fold increase on the node), memory size (roughly 20-fold decrease in ratio of memory to flops) and speed and at least constant performance relative to peak [104], and limited I/O. New computer architectures will be required and the expected evolution of current commodity computing hard-

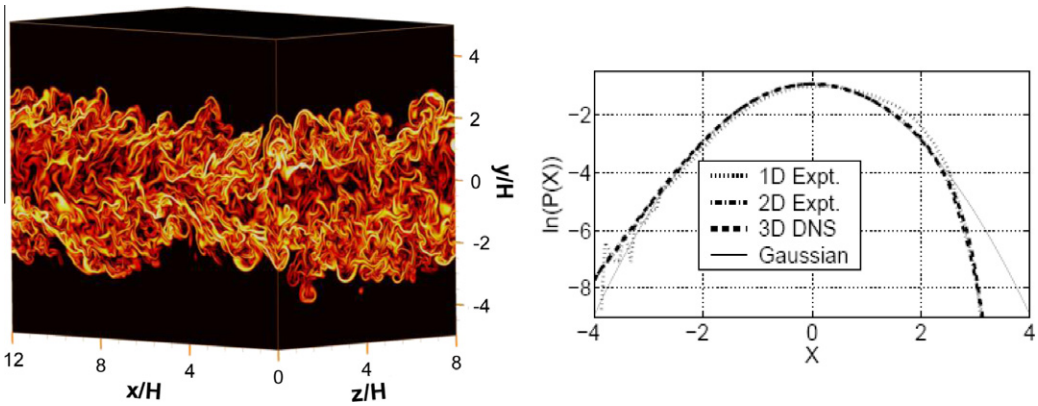


Fig. 16. (Left image) Configuration diagram showing contours of $\ln(\chi_3/\chi_q)$, where $\chi_q$ is the steady extinction dissipation rate $2194s^{-1}$, at time $30t_j$ [10]. 1D $\chi$ measurements were evaluated in the $y$ and $z$ directions, and a 2D measurement evaluated in the $x$–$y$ plane. (right image) PDFS of $\chi = (\ln(\chi_T) - \mu_T)/\sigma_T$ reconstructed from 1D radial (1D Expt.) and 2D radial–axial (2D Expt.) experimental measurement [98,99], normalized by the 2D reconstructed mean and rms; 3D DNS PDF, normalized by its mean and rms (3D DNS); a Gaussian PDF (Gaussian) (Fig. 9 from [93]).

ware will not get us there. Software – encompassing everything from systems software to new programming models to new energy aware and fault resilient algorithms for solving large systems of partial differential equations – for exascale computers will require new approaches to efficiently use such a massive number of processors. Concurrent design, or 'co-design' [105–107] of everything from hardware to the entire software stack including new algorithms and DNS software is required if computational combustion scientists are to make effective use of exascale computers. This will require unprecedented collaboration between computer scientists, applied mathematicians and combustion scientists well in advance of the anticipated deployment of exascale machines within the next decade.

Most application codes on today's homogeneous cpu-based distributed memory computers implement a standard single-program multiple data (SPMD) program using the message passing interface MPI for communication to achieve scaleable parallelism. To achieve performance at the exascale on hybrid architectures, critical existing algorithms in DNS of combustion will have to be entirely redesigned, trading flops for memory and off-node communication. Inter-node communication will likely continue to use the MPI for scaleable parallelism, but intra-node performance will likely be achieved through some optimum combination of OpenMP and CUDA or other modern programming languages adapted to conventional acceleration frameworks, e.g. Cell Broadband Engine (CBE), GPU, and SSE. Accelerator architectures typically require coalesced threads and vectorization to achieve good performance. Existing DNS data structures which are typically 'array of structures' (AOS), with all variables at a single grid point laid out contiguously in memory, are unlikely to perform well on accelerators. Rearrangement of the internal data structures of existing DNS codes to use 'structure of arrays' (SOA), with a single arbitrary direction contiguous in memory will facilitate vectorization of all point-wise operations. Future hybrid arhitectures are also likely to be comprised of heterogeneous compute notes with FLOP-rich accelerators (GPU, CBE, and SSE) connected via a slow link like a PCIe bus. Hence, where possible, it may be advantageous to minimize data movement between key kernels in the code, and therefore, if feasible as much of the computation for the entire time step should be moved to the accelerator, with only the remaining I/O and inter-node communication performed on the host. Other challenges stem from the imbalance between the serial speed-up on the accelerator and the host cpu compute performance, resulting in an imbalance between compute and communication. Therefore, it will be necessary to reorganize the communication patterns to minimize latency and data movement, and to increase local computation while minimizing inter-node communication. Recognizing these needed algorithmic changes, i.e. more computation and less data movement, has recently led to the re-design of a tri-diagonal Thomas solver as part of a compact Pade finite-difference scheme in a compressible flow DNS solver [108]. As a result of the refactorization the DNS code was sped up by 20-fold compared to the cpu-based code. Further strategies to mask communication overhead by overlapping local compute with data movement between nodes and minimizing synchronization points between the host and accelerator also need to be explored to enhance the overall performance.

Combustion DNS on exascale systems will also produce unprecedented amounts of data that

must be indexed, analyzed, and reduced in situ, prior to storage or subsequent analysis because data movement on such systems and I/O will be severely constrained. This represents a paradigm shift in the post-analysis and visualization of large datasets. In situ data reduction methods will become essential to identify pertinent information for subsequent analysis and to reduce the amount of data written to disk. To reduce the amount of data that has to be written to disk and transferred, typically restart files at predetermined time-steps are stored on disk, and consequently, only coarse-grained temporal resolution of the raw data is available for subsequent time-series analysis and visualization of the data. This situation is not optimal for understanding intermittent combustion phenomena like extinction or reignition, whose occurence is unknown a priori and hence, unlikely at the intervals that the data is stored, and also cumbersome as larger amounts of data from petascale DNS are generated and need to be written to disk. It is advantageous to process the data while the DNS is being performed, so that the amount of data that has to be stored or transferred is kept to a minimum.

Recently, in situ parallel data processing and visualization strategies in a massively parallel combustion DNS environment were investigated. Detailed testing demonstrate that in situ visualization is a promising method for accelerating the overall process of high-performance supercomputing and scientific discovery. Several outstanding issues related to in situ processing were identified and addressed: (1) the visualization and analysis code has to interact directly with the DNS code. For optimal memory usage, the simulation and visualization codes share the same data structures as the DNS code to avoid data replication; (2) as a consequence, the visualization workload balancing is harder to achieve since the visualization has to comply with the DNS architecture, i.e. data partitioning and distribution is dictated by the DNS code. Hence, there is a need to balance the visualization workload so the visualization and analysis aspects are scaleable; and (3) the visualization costs need to be kept low so that their overall cost is only a small fraction ($<$ 1%) of the overall computational cost of the DNS. Data partitioning, parallel rendering and parallel image compositing implementation issues are addressed in [109] and evaluation of an in situ visualization capability for combustion DNS shows good performance on a CrayXT5 up to approximately 15,000 cores tested.

Feature extraction, classification and tracking is another example of a useful strategy for data reduction. Regions of interest in a turbulent flow field and its relationship with intermittent combustion events, e.g. extinction and ignition, that can be readily characterized by local minima or maxima can be identified and tracked using topological methods. Contour trees [110–113], Morse-Smale complexes [114–117], Jacobi sets [118], Hodge decompositions [119,120], and topological terrains [121] are some of the topological analysis techniques that can be used to identify, describe, and track salient features and to guide subsequent statistical analysis used to glean insight from large combustion simulations. Recently, topological feature segmentation and tracking methods have been applied to understanding the evolution of turbulent combustion phenomena including the evolution of ignition kernels, mixture fraction scalar dissipation rate, and cellular flame structure [122–126]. Topological or geometric methods provide a rigorous mathematical framework for non-local analysis of fluid, mixing, and combustion structure. Many algorithmic and computer science challenges still remain in the efficient, scaleable parallel implementation of these methods in situ on exascale architectures. Recently, a distributed parallel algorithm was developed to compute a merge tree for analyzing massive combustion DNS data [127]. The merge tree encodes the merging behavior of level sets (isosurfaces) as a sweep is performed from the maximum level down to the minimum in the computational domain. Topological features of level sets, such as connected components, handles, and voids, can be important aids in interpreting the data. The merge tree captures the merging behavior of the connected components of level sets. The parallel merge tree algorithm was implemented on up to 130K processes on a CrayXT5 and obtained running times in the tens of seconds for a 1.3 billion point multi-variate combustion DNS data-set. In timing measurements, some parts of the algorithm are fast – in particular the per process local merge tree computation – and therefore good candidates for in situ processing, while the combine phase of the algorithm is slower and may have to be offloaded to a co-processing system to not slow down the DNS run.

### 3.2. Opportunities for exascale combustion DNS

If DNS codes can successfully utilize future exascale architectures, DNS has the potential to transform our understanding of the coupling between low-temperature ignition kinetics and turbulent mixing at high pressure that determines lifted flame stabilization, ignition timing, rate of combustion, and emissions characteristics in future engines and gas turbines. Understanding complex low-temperature high pressure kinetics of alternative fuels and its coupling with turbulent transport at high pressure requires much greater resolution and the transport of large numbers of reactive scalars enabled by exascale computing power. Moreover, in situ adaptive reduction strategies for accurate and computationally affordable inclusion of chemical kinetics with quantified

uncertainties in DNS will expedite DNS of turbulent combustion. The insights gained from these simulations will enable the development of predictive multi-scale models to optimally design future evolving fuels and engines.

A leading candidate of advanced internal combustion engine concepts are low temperature, high pressure, compression ignition engines. This is the only engine strategy that can meet future government-mandated standards on decreased $NO_x$/soot emissions and increased fuel efficiency. However, prototype engines of this design are hard to control because of sensitivity to cycle-to-cycle variations inside of the piston cylinder. These variations arise from the variability of the low temperature and the spark-less compression ignition, both of which are dependent on turbulent mixture preparation and details of the fuel chemistry. Simulation currently can play only a minor role in influencing engine design because the coupling of turbulence with detailed fuel chemistry over multiple engine cycles is beyond both our scientific understanding and current computational resources. Three-dimensional DNS of turbulent combustion at the micro-scale in compression ignition engine environments will aid in understanding the underlying physical mechanisms controlling spontaneous autoignition of mixture pockets subjected to varying degrees of thermal and compositional inhomogeneities and provide essential statistical information needed to develop and validate models used in engineering RANS and LES simulations.

In summary, future DNS running on exascale computing systems will enable deep understanding of underlying chemical and combustion science processes, complementing experimentation and theory. This will enable reduced parameterizations to be developed and incorporated in engineering CFD to enhance combustion device design and performance, and ultimately greatly reduce engine and gas turbine development timescales and costs, while meeting energy security and emissions regulations in a timely manner, and enhancing competitiveness in key technologies.

## Acknowledgments

This work was supported by the Division of Chemical Sciences, Geosciences and Biosciences, the Office of Basic Energy Sciences, the US Department of Energy (DOE) and computer allocations were awarded by DOE's Innovative and Novel Computational Impact on Theory and Experiments (INCITE) program. This research used resources of the National Energy Research Computing Center (NERSC), of the National Center for Computational Sciences at Oak Ridge National Laboratory (NCCS/ORNL) which are supported by the Office of Science of the US DOE under Contract Nos. DE-AC03-76SF00098 and DE-AC05-00OR22725, respectively. The author wishes to acknowledge the contributions of E.R. Hawkes, O. Chatakonda, E. Knudsen, M. Tanahashi, S. Cant, L. Vervisch, J. Bell, M. Day, T. Lu, C.S. Yoo, E.S. Richardson, A. Mascarenhas, R.W. Grout, and H. Pitsch to this paper.

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